

DAY 8

EDA

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[2]: df=sns.load_dataset("tips")
```

```
[30]: print(df.head(20))
```

```
total_bill  tip    sex smoker  day  time  size
0      16.99   1.01  Female    No  Sun  Dinner    2
1      10.34   1.66    Male    No  Sun  Dinner    3
2      21.01   3.50    Male    No  Sun  Dinner    3
3      23.68   3.31    Male    No  Sun  Dinner    2
4      24.59   3.61  Female    No  Sun  Dinner    4
5      25.29   4.71    Male    No  Sun  Dinner    4
6       8.77   2.00    Male    No  Sun  Dinner    2
7      26.88   3.12    Male    No  Sun  Dinner    4
8      15.04   1.96    Male    No  Sun  Dinner    2
9      14.78   3.23    Male    No  Sun  Dinner    2
10     10.27   1.71    Male    No  Sun  Dinner    2
11     35.26   5.00  Female    No  Sun  Dinner    4
12     15.42   1.57    Male    No  Sun  Dinner    2
13     18.43   3.00    Male    No  Sun  Dinner    4
14     14.83   3.02  Female    No  Sun  Dinner    2
15     21.58   3.92    Male    No  Sun  Dinner    2
16     10.33   1.67  Female    No  Sun  Dinner    3
17     16.29   3.71    Male    No  Sun  Dinner    3
18     16.97   3.50  Female    No  Sun  Dinner    3
19     20.65   3.35    Male    No  Sat  Dinner    3
```

```
[4]: print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   total_bill  244 non-null    float64
1   tip         244 non-null    float64
2   sex         244 non-null    category
```

```
[5]: print(df.columns)
```

```
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='object')
```

```
[6]: print(df.describe())
```

```
total_bill  tip    size
count  244.000000  244.000000  244.000000
mean    19.785943   2.998279   2.569672
std      8.902412   1.383638   0.951100
min      3.070000   1.000000   1.000000
25%     13.347500   2.000000   2.000000
50%     17.795000   2.900000   2.000000
75%     24.127500   3.562500   3.000000
max      50.810000  10.000000   6.000000
```

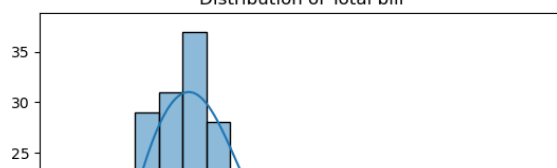
```
[10]: #Descriptive Statistics
print(df['tip'].describe()) #for a particular column
```

```
count    244.000000
mean      2.998279
std       1.383638
min       1.000000
25%       2.000000
50%       2.900000
75%       3.562500
max       10.000000
Name: tip, dtype: float64
```

```
[14]: #HISTOGRAM
```

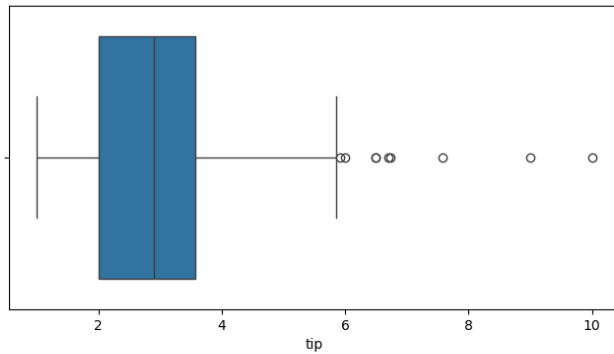
```
sns.histplot(df['total_bill'],bins=20,kde=True)
plt.title("Distribution of Total bill")
plt.show()
```

Distribution of Total bill



```
[16]: plt.figure(figsize=(8,4))
sns.boxplot(x=df['tip']) #finding outliers for tip column
```

```
[16]: <Axes: xlabel='tip'>
```



Categorical Value

```
[18]: #VALUE COUNT OF DAY
print(df['day'].value_counts())
```

```
day
Sat    87
Sun    76
Thur    62
Fri     19
Name: count, dtype: int64
```

Jupyter day8 Last Checkpoint: 2 days ago

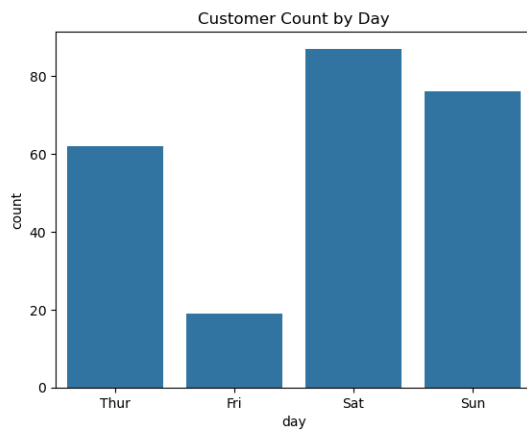
File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python [conda env:base] * Anaconda Toolbox

```
Name: count, dtype: int64
```

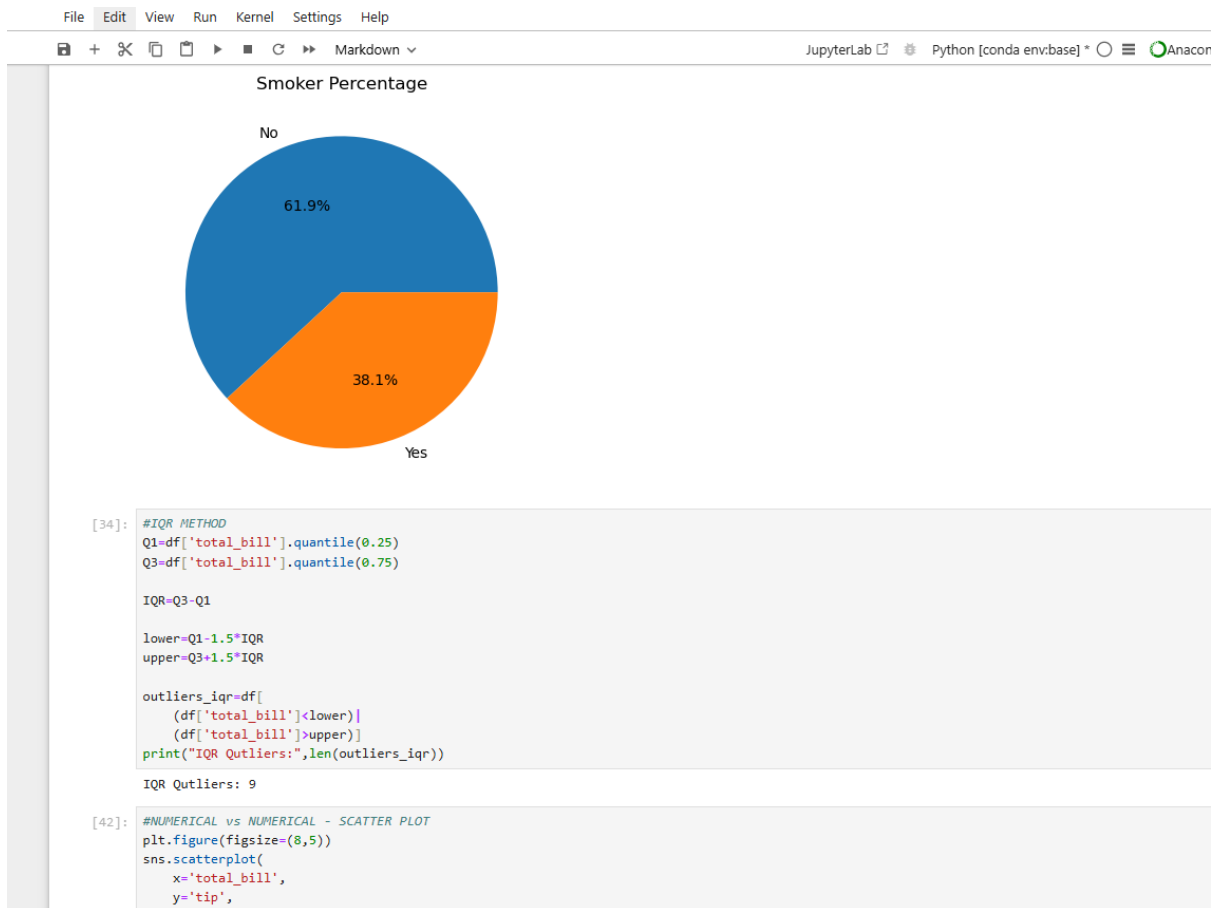
```
[20]: #COUNT PLOT
sns.countplot(x='day',data=df)
plt.title("Customer Count by Day")
plt.show()
```



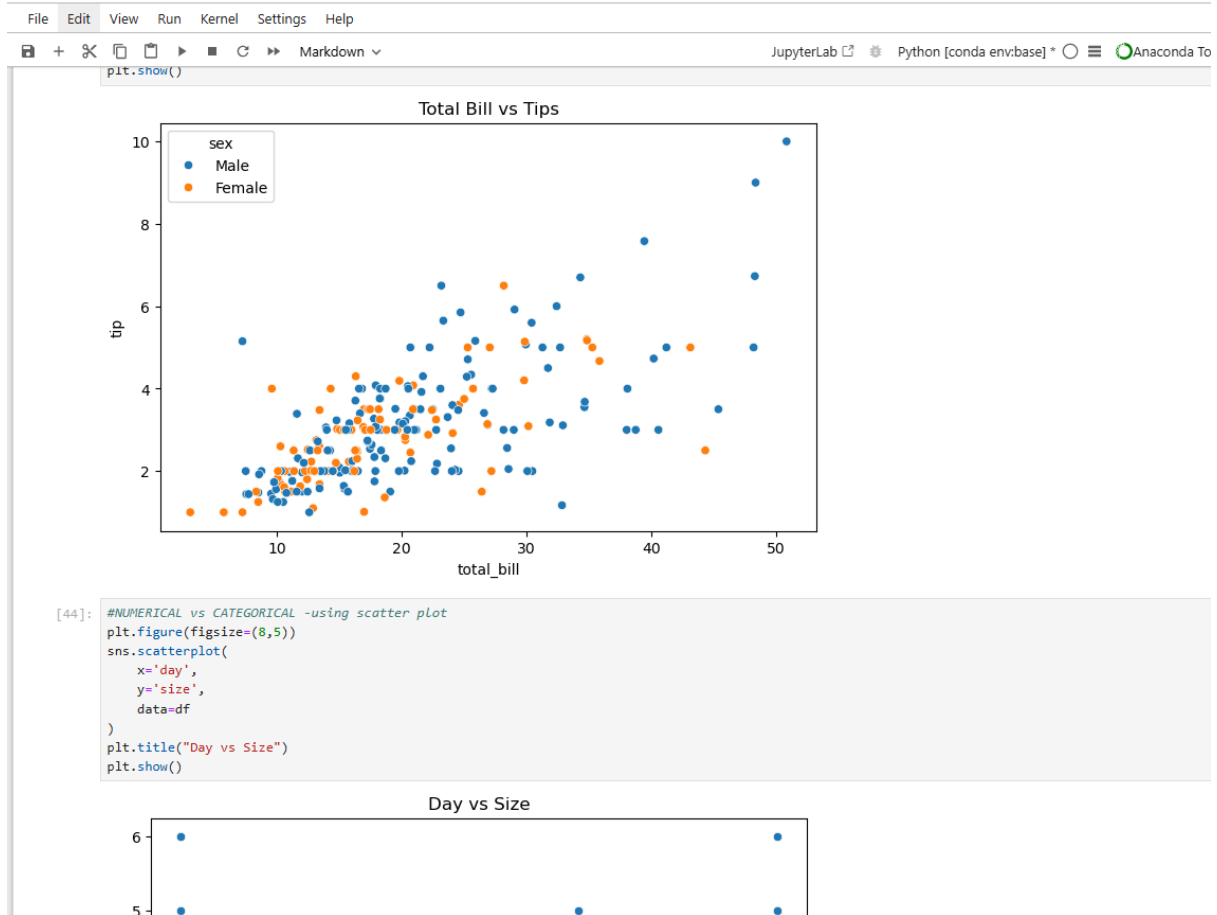
```
[33]: #PIE CHART
df['smoker'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%' #this line is used to display the percent value in the center ... ( 61.9% and 38.1%)
)
plt.title("Smoker Percentage")
plt.ylabel("")
plt.show()
```

Smoker Percentage

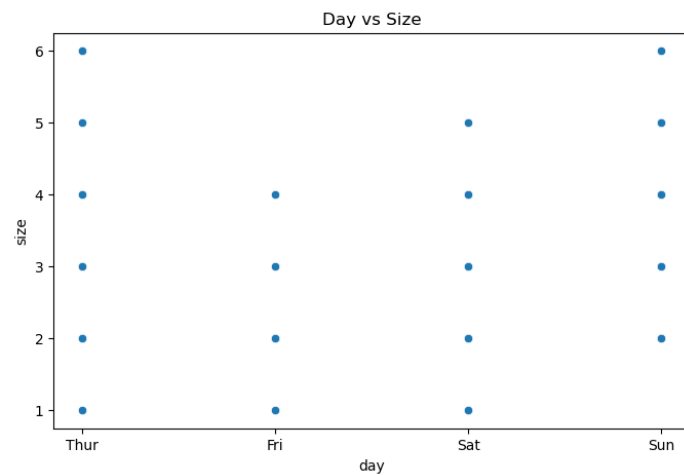
No



Jupyter day8 Last Checkpoint: 2 days ago



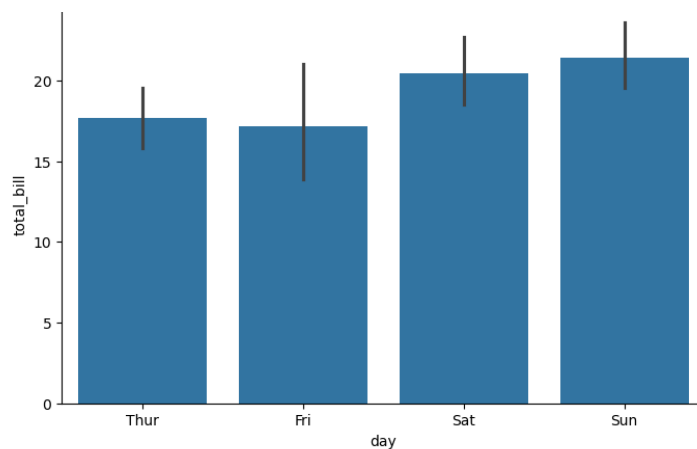
plt.show()



```
[45]: #num vs cat val
#Using correlation - Pearson Correlation
corr=df['total_bill'].corr(df['tip'])
print(corr)

0.6757341092113641
```

```
[49]: #BAR CHART
plt.figure(figsize=(8,5))
sns.barplot(
    x='day',
    y='total_bill',
    data=df
)
plt.title("Average Total Bill by Day")
plt.show()
```



```
[53]: #GROUPED STATISTICS
#MEAN BILL BY DAY
print(
    df.groupby('day',observed=False)['total_bill'].mean()
)

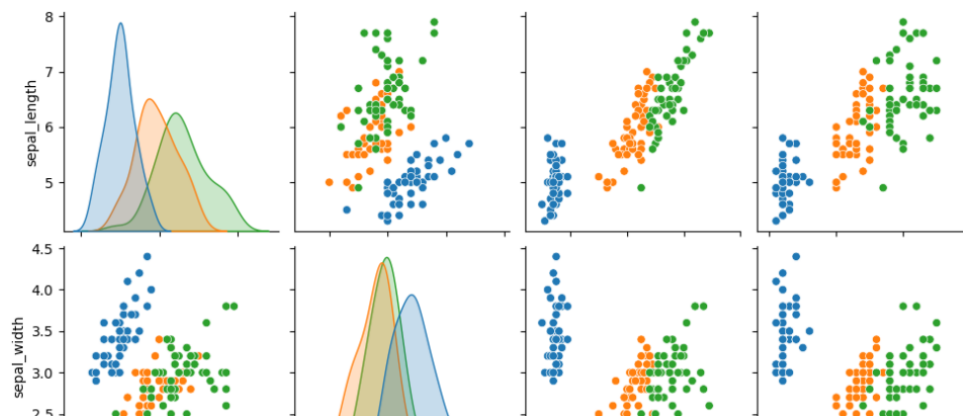
# multiple statistics
print(
    df.groupby('day',observed=False)['total_bill'].agg(['mean','median','min','max'])
)

day
Thur    17.682742
Fri     17.151579
Sat     20.441379
Sun     21.410000
Name: total_bill, dtype: float64
      mean median  min  max
day
```

```
[ ]: #CROSSTAB
#SEX vs SMOKER
pd.crosstab(
    df['sex'],
    df['smoker']
).plot(kind='bar')
plt.title("Sex vs Smoker")
plt.show()
```

Multivariate analysis

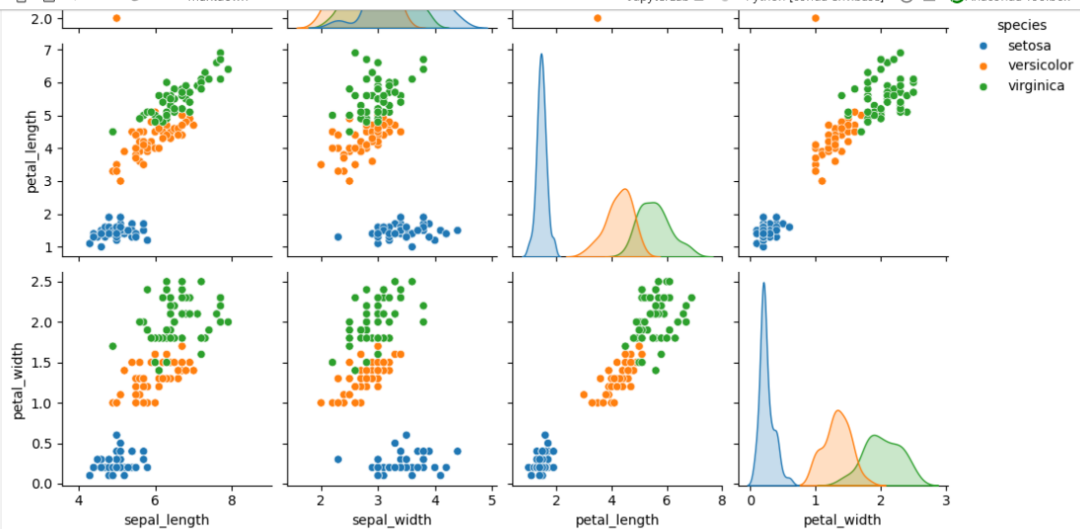
```
[62]: #PAIR PLOT
import seaborn as sns
import matplotlib.pyplot as plt
df=sns.load_dataset("iris")
sns.pairplot(
    df,
    hue='species'
)
plt.show()
```



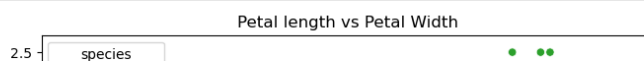
File Edit View Run Kernel Settings Help

Trusted

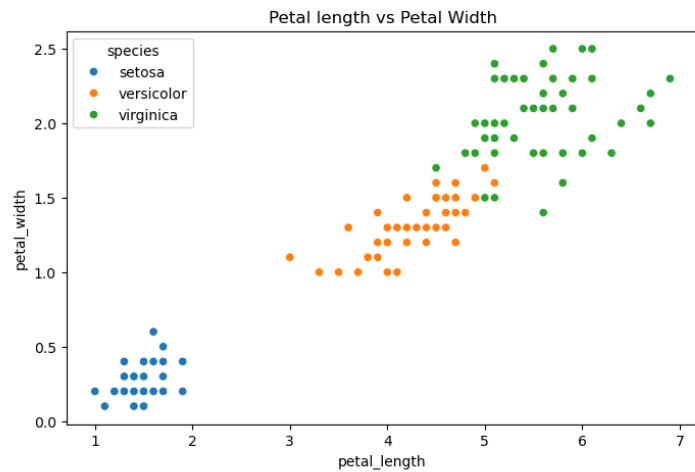
Python [conda env:base] * Anaconda Toolbox



```
[64]: #COLOR CODED SCATTER PLOT (3-DIMENSION)
plt.figure(figsize=(8,5))
sns.scatterplot(
    x='petal_length',
    y='petal_width',
    hue='species',
    data=df
)
plt.title("Petal length vs Petal Width")
plt.show()
```



```
plt.show()
```



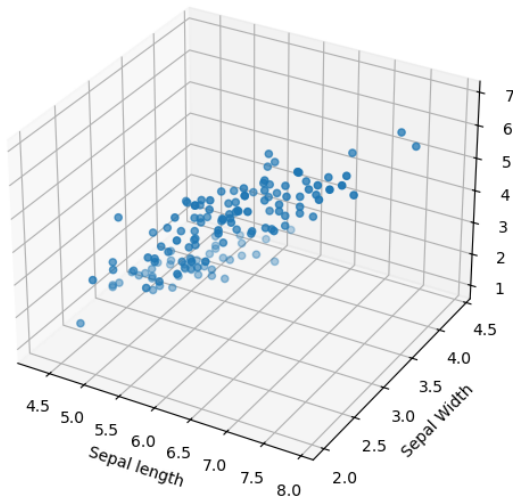
```
[67]: #3-D SCATTER PLOT
```

```
from mpl_toolkits.mplot3d import Axes3D
import matplotlib.pyplot as plt

fig=plt.figure(figsize=(8,6))
ax=fig.add_subplot(111,projection='3d')
ax.scatter(
    df['sepal_length'],
    df['sepal_width'],
    df['petal_length']
)
ax.set_xlabel("Sepal length")
ax.set_ylabel("Sepal Width")
ax.set_zlabel("Petal Length")
plt.title(" 3D Projection")

plt.show()
```

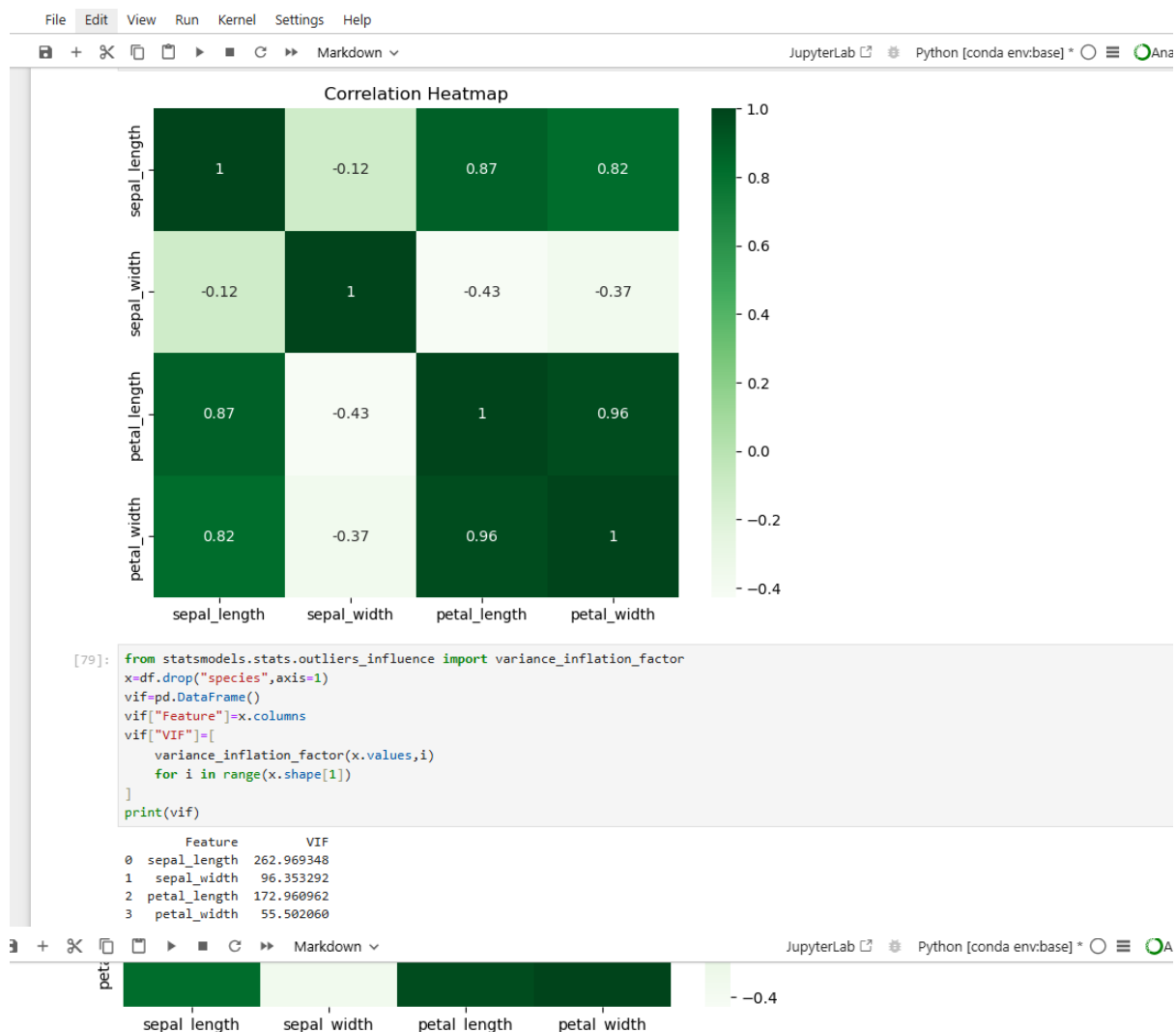
3D Projection



```
[68]: corr_matrix=df.corr(numeric_only=True)
print(corr_matrix)
```

	sepal_length	sepal_width	petal_length	petal_width
sepal_length	1.000000	-0.117570	0.871754	0.817941
sepal_width	-0.117570	1.000000	-0.428440	-0.366126
petal_length	0.871754	-0.428440	1.000000	0.962865
petal_width	0.817941	-0.366126	0.962865	1.000000

```
[75]: plt.figure(figsize=(8,6))
sns.heatmap(
    df.corr(numeric_only=True),
    annot=True,
    cmap='Greens'
)
```



```
[79]: from statsmodels.stats.outliers_influence import variance_inflation_factor
x=df.drop("species",axis=1)
vif=pd.DataFrame()
vif["Feature"]=x.columns
vif["VIF"]=[
    variance_inflation_factor(x.values,i)
    for i in range(x.shape[1])
]
print(vif)
```

	Feature	VIF
0	sepal_length	262.969348
1	sepal_width	96.353292
2	petal_length	172.960962
3	petal_width	55.502060

```
[83]: #REMOVE A HIGH-VIF FEATURE
#suppose we remove petal_length
x_new=x.drop("sepal_length",axis=1)
vif_after=pd.DataFrame()
vif_after["Feature"]=x_new.columns
vif_after["VIF"]=[
    variance_inflation_factor(x_new.values,i)
    for i in range(x_new.shape[1])
]
print(vif_after)
```

	Feature	VIF
0	sepal_width	5.856965
1	petal_length	62.071308
2	petal_width	43.292574

```
[ ]:
```